



Network Traffic Monitor

Real-time network traffic monitor for Windows

[User Manual](#)

Version 0.1.0

Designed by Codecider Lab

1. Overview

Network Traffic Monitor is a Windows desktop application that captures and visualizes live network traffic in real time. The left panel shows an at-a-glance summary (speeds, chart, statistics, top talkers, alerts); the right panel shows a detailed, filterable packet log.

It analyzes L3/L4 metadata for TCP, UDP, DNS, ICMP, IPv4, IPv6 and ARP. It does **not** decrypt HTTPS content or block packets.

2. System Requirements

Item	Description
Operating System	Windows 10 or Windows 11
Python	3.12 or later

Capture driver	Npcap (required for real capture)
Privileges	Administrator (for real capture)
Libraries	PySide6, pyqtgraph, scapy, psutil

3. Installation

1. Install Python 3.12+ and add it to PATH.
2. Install Npcap from <https://npcap.com> and enable the "WinPcap API-compatible mode" option.
3. Install dependencies (a virtual environment is created automatically):

```
pip install -r requirements.txt
```

4. Running the Application

The simplest way to launch is the bundled batch file:

```
run.bat    :: real capture (requests admin rights)
```

```
run.bat --demo  :: preview UI with synthetic traffic
```

On first run, **run.bat** creates a virtual environment and installs dependencies. Real-capture mode automatically requests Administrator elevation. Demo mode needs neither Npcap nor admin rights and is ideal for exploring the UI.

You can also start it directly:

```
python -m network_monitor.main --demo
```

5. Interface Overview

Area	What it shows
Toolbar	Interface selector, refresh, Start / Pause / Stop, SQLite toggle, live capture status.
Summary panel (left)	Download/upload speed, 60-second chart, protocol statistics, top processes and destinations, alerts.
Log table (right)	Real-time per-packet log with color-coded direction and protocol.
Filter bar	Display filters applied to the visible log.
Action bar	Auto-scroll toggle, row copy, clear, CSV / JSONL export.
Status bar	Current status messages and the product credit.

6. Capturing Traffic

- Select a network interface from the toolbar dropdown (use the refresh button to re-scan).
- Click **Start** to begin capturing on the selected interface.
- Click **Pause** to freeze the log while capture and statistics continue; click again to resume.
- Click **Stop** to end capture and cleanly shut down all worker threads.

Capture, parsing, and storage run on background threads, so the interface stays responsive even under heavy traffic. If the packet queue saturates, the dropped count is shown rather than freezing the app.

7. Reading the Summary Panel

Element	Meaning
Download / Upload	Current inbound / outbound speed with automatic units.
60-second chart	Rolling download (green) and upload (blue) speed history.
Statistics	Active connections, packets/s, and per-protocol counts.
Top Processes	Processes using the most bandwidth.

Top Destinations	Remote hosts receiving the most bytes.
Alerts	Threshold warnings such as high upload or dropped packets.

8. The Traffic Log Table

Column	Description
Time	Packet capture time.
Dir	IN (inbound), OUT (outbound), or LOCAL.
Proto	TCP, UDP, DNS, ICMP, or ARP.
Process / PID	Owning process name and id (when resolvable).
Source / S.Port	Source IP address and port.
Destination / D.Port	Destination IP (or domain) and port.
Size	Packet length.
Summary	Short human-readable description.

Double-click any row to open the packet detail view.

9. Filtering

The filter bar applies a **display filter**: it changes what is shown without discarding captured data. Clear the fields to see everything again.

- Protocol and Direction dropdowns.
- IP address, Port, and Process substring search.
- Free-text search across source, destination, protocol, domain, and summary.

A **capture filter** (BPF, e.g. *tcp port 443*) limits which packets are captured in the first place and is set via configuration.

10. Packet Detail View

Double-clicking a log row opens a dialog with the full metadata (time, direction, interface, protocol, process, addresses, ports, length, summary) and a Hex / ASCII dump of the raw bytes.

For privacy, raw payload capture is **off by default**; enable *store_raw_payload* in configuration to populate the hex view.

11. Saving and Exporting

Format	How to use
SQLite	Enable the toolbar toggle to persist packets to a database (batched, non-blocking).
CSV	Export the current log via the action bar.
JSON Lines	Export the current log as one JSON object per line.

12. Configuration

Settings are read from an optional TOML file (pass it with `--config`):

```
[capture]
default_interface = ""
bpf_filter = "ip or ip6"
queue_size = 50000
```

```
store_raw_payload = false

[ui]
refresh_interval_ms = 250
chart_interval_ms = 1000
max_visible_rows = 10000
auto_scroll = true

[storage]
enable_sqlite = false
database_path = "./data/traffic.db"
batch_size = 100
flush_interval_ms = 1000
```

13. Troubleshooting

Symptom	Resolution
"Npcap is not installed"	Install Npcap from npcap.com with WinPcap-compatible mode, then restart. Use --demo to preview without capture.
No interfaces listed	Run as Administrator and click the refresh button.
Process names missing	Some short-lived or privileged connections cannot be mapped; they show as empty.
Dropped packets shown	Traffic exceeded processing capacity; the app keeps running and reports the count.
UI text looks garbled in console	Console-only encoding issue (e.g. localized adapter names); the GUI and log file are correct.

14. Privacy and Security

- Captured metadata can be sensitive; raw payloads are not stored by default.
- The application never uploads data or performs remote analytics.
- Review exported files before sharing, as they may contain private information.
- Administrator rights are used only for packet capture.

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